

MAR: Metacognitive Agentic Reasoning for Multimodal Fake News Detection

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Abstract

Multimodal fake news combining text and images has become increasingly prevalent, fueled by the rapid dissemination on social media. Existing approaches predominantly rely on supervised learning–driven small multimodal language models, yet they are constrained by the knowledge scope and logical reasoning capabilities limited by training data: their performance degrades significantly when encountering novel content not covered in the training data, and they typically provide only uninterpretable binary true/false predictions. To address these challenges, we propose the **M**etacognitive **A**gentic **R**easoning for Multimodal Fake News Detection(**MAR**), a framework that integrates multi-agent metacognitive debate with external knowledge retrieval to enhance the accuracy, generalization, and interpretability of multimodal fake news detection. At its core, MAR introduces a news-domain informed and metacognition-inspired multi-agent reasoning mechanism: it first generates several prior pseudo-labels of the news domain, and defines the characteristics of two opposing agents (i.e., Believer and Skeptic) leveraging these pseudo-labels; then, the Believer and Skeptic go through a three-stage "Draft – Self-Critique – Refine" interactive debate simulating humans' learning behavior, called the metacognitive debate. Specifically, MAR first generates an initial evidence-grounded judgment (Draft), then critically reflects on their own and others' arguments (Self-Critique), and finally iteratively refines their conclusions by incorporating internal or external evidence (Refine). To improve factual reliability, the framework dynamically retrieves trustworthy external knowledge via web and reverse image searches, thereby mitigating the hallucinations inherent in large language models. Experiments show that MAR achieves state-of-the-art performance on two benchmarks and significantly outperforms existing methods in terms of generalization and interpretability. The source code is available at https://github.com/Averdgr/MAR_KDD.

CCS Concepts

• **Information systems** → *Data mining*; • **Computing methodologies** → *Natural language processing*.

Keywords

Fake News Detection, Multi-Modal, Large Vision Language Models, Fact-Checking

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Resource Availability:
The source code of this paper has been made publicly available at https://doi.org/10.5281/zenodo.20485147.

1 Introduction

With the rapid development of social media and digital platforms, especially the widespread adoption of AIGC technologies[9, 47], the barrier to producing fake news has been lowered[34, 45]. This has turned fake news into a critical issue, threatening the information ecosystem[28], misleading public understanding[7], and

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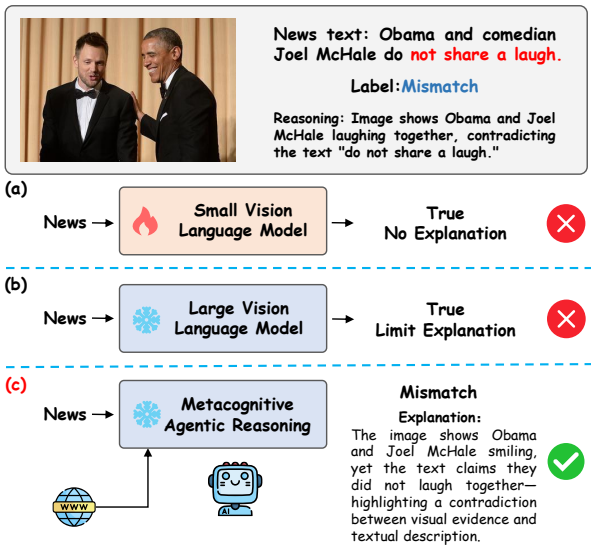


Figure 1: Illustration of the roles of SVLMs and LVLMs and our proposed Metacognitive Agentic Reasoning. (a) SVLMs alone lack interpretability; (b) LVLMs still misclassifies and offers limited explanations; (c) Metacognitive Agentic Reasoning correctly identifies text-image mismatch and provides accurate explanations.

even affecting social stability[3]. For instance, ahead of the 2024 U.S. presidential election, a surge in fake news spread rapidly across digital platforms, severely undermining public trust in information sources.[1]

Early approaches to fake news detection mainly relied on textual analysis to assess authenticity by examining linguistic features, semantic coherence, or dissemination patterns[26]. However, in the current information consumption environment, which is dominated by visual content, fake news increasingly appears in multimodal forms that combine images and text[4]. Such fake news often employs images generated or manipulated by artificial intelligence, together with carefully crafted misleading text, to create a perception that "seeing is believing," thereby greatly enhancing its deceptive power and capacity for dissemination[17]. Multimodal Fake News Detection (MFND) aims to integrate information from text and images across multiple dimensions[20, 40], including semantics, emotion, factual consistency, and external knowledge[30], in order to jointly identify contradictions, misleading cues, or fabricated signals, and thus accurately determine whether an image-text pair constitutes fake news[11].

Most existing multimodal fake news detection methods still rely on supervised, task specific small scale multimodal language models[23, 48]. However, constrained by the limited size and coverage of training datasets, as well as the insufficient knowledge and logical reasoning capabilities of the models themselves, these methods tend to perform well only on their respective datasets, while their performance often degrades significantly when confronted with novel news topics or entities not covered in the training data[29]. In addition, such models usually provide only a final true/false judgment, lacking the ability to explain the rationale behind their decisions to users.

With the development of large language models (LLMs), LLM-based fake news detection methods have demonstrated outstanding performance in multimodal tasks[1, 22]. Inspired by research on multi-agent system simulation, multi-agent debate has attracted widespread attention due to its ability to simulate adversarial reasoning processes[10]. This approach deploys multiple LLM agents to conduct structured argumentation, aiming to reveal factual consistency through the confrontation of differing viewpoints. Systems composed of multiple agents can exploit the complementary strengths of competing perspectives, thereby producing more robust judgments than those generated by a single classifier[37]. Nevertheless, existing research based on multi-agent debate exhibits notable limitations. Early studies primarily relied on the internal knowledge of LLMs and confined the debate process to the textual modality[15, 25]. This practice not only ignores visual information but also leads agents into ineffective arguments driven by the intrinsic hallucination problem of LLMs, resulting in debates that prioritize rhetorical persuasion over factual grounding.

More importantly, mainstream multimodal fake news detection methods remain coarse-grained in terms of output. Their predictions are either limited to binary labels of truthfulness[41] or confined to identifying whether images and text are out-of-context[31]. In reality, multimodal fake news detection is inherently a fine-grained and complex task that requires the simultaneous identification of multiple types of manipulation, including textual distortion, visual distortion, and cross-modal distortion[24]. The core challenge lies in the fact that fake news exhibits highly diverse and covert manipulation patterns in multimodal content. It is not merely manifested as a binary opposition between true and false at the global level, but more often achieves its deceptive goals through localized, subtle, and cross-modal manipulations.

In this context, we revisit the proverb: "Rumors stop at the wise". What makes a "wise" individual capable of stopping rumors? Not merely access to facts, but the ability to critically reflect, seek external evidence, and revise beliefs iteratively in light of new information—processes rooted in metacognition: thinking about thinking.

Inspired by this human-like epistemic behavior, we propose the Metacognitive Agentic Reasoning for Multimodal Fake News Detection(MAR), a novel evidence-driven framework that simulates the reasoning process of a collective of wise agents. MAR integrates a multi-agent debate mechanism, in which each agent performs a "Draft – Self-Critique – Refine" cycle to mimic human metacognitive behavior through initial judgment, introspective critique, and iterative improvement, together with a external knowledge retrieval module that is selectively triggered during debate rounds to retrieve and filter relevant evidence from web search and reverse image lookup, thereby grounding the agents' reasoning and mitigating hallucinations. By combining competitive dialogue with reflective revision and fact grounding, MAR not only achieves high accuracy and interpretability but also provides a traceable reasoning chain for each decision. Our main contributions are as follows:

[1]https://www.bbc.com/news/articles/czj7eex29r3o

- **Evidence-Driven Framework**: We propose MAR, a comprehensive evidence-driven framework for multimodal fake news detection. At its core, MAR leverages a synergistic interplay between External Knowledge Retrieval and Metacognitive Debate, forming a dynamic closed-loop to anchor model reasoning in factual evidence and mitigate hallucinations.
- **Fact Verification & Metacognitive Debate**: In our external fact verification pipeline, web search results are first coarsely filtered using BGE and then refined by LVLM to eliminate noise; reverse image search further retrieves evidence from news images. The debate module adopts a three-stage "Draft – Self-Critique – Refine" reasoning process to simulate human metacognition, enabling more rational debates.
- **Effectiveness**: Experiments on two fine-grained multimodal datasets show that MAR outperforms existing methods in accuracy, generalization, and interpretability, while providing a traceable reasoning chain.

2 Related Work

2.1 Multimodal Fake News Detection

Multimodal fake news detection can be categorized into several sub-tasks based on the type of forgery, primarily including textual distortion detection, visual distortion detection, and cross-modal distortion detection[24]. In terms of textual distortion detection, existing studies largely rely on language models to determine authenticity solely based on textual content[29, 38]. Moreover, many approaches further incorporate auxiliary features, such as user comments[33, 46] and social graphs[27, 42], to enhance prediction performance. However, these methods largely overlook visual evidence, which is often crucial for reliable fact verification[2]. Regarding visual distortion detection, researchers have attempted to enhance discrimination capabilities by injecting forgery-related knowledge into Large Vision-Language Models (LVLMs)[23] or by incorporating reinforcement learning to achieve detection of forged images[18]. Addressing cross-modal inconsistencies, mainstream methods focus on identifying semantic or entity-level mismatches between news text and accompanying images[13, 36], often introducing external knowledge bases to improve fact-checking accuracy[1, 22, 31].

Recently, research has shifted focus toward more challenging complex scenarios that encompass multiple concurrent forgery types (e.g., text distortion, image distortion, and cross-modal distortion). MMD-Agent[24] constructs a multi-agent system that explicitly decomposes the detection process into the three aforementioned sub-tasks through step-by-step reasoning and external knowledge augmentation. LRQ-FACT[8] utilizes Large Language Models (LLMs) to automatically generate text- and image-oriented Fact-Checking Questions (FCQs), guiding the retrieval and verification of multimodal evidence. T^2Agent[12] designs a tool-augmented multi-agent architecture, incorporating Monte Carlo Tree Search (MCTS) to enable the synergistic verification of multi-source evidence. However, these methods predominantly adopt multi-stage or modular strategies to handle distinct forgery types separately. Consequently, they lack the capacity for joint modeling and optimization of multimodal information, making it difficult to fully capture the deep semantic correlations across modalities.

2.2 Multi-Agent Debate

Multi-agent debate (MAD) simulates a structured debating process that leverages a group of agents to fully utilize their individual capabilities and expertise, thereby enhancing the efficiency and effectiveness of handling complex tasks[10]. For detecting multimodal fake news characterized by inconsistencies between text and images, researchers have designed multi-agent debate systems to identify contextual discrepancies between the two modalities[21]. Notably, some scholars have built structured multi-agent debate frameworks[14, 25] powered by large language models, significantly improving the interpretability of detection outcomes. This interpretability is further strengthened through the integration of multidimensional evaluation mechanisms[15] that support more transparent and reasoned decisions. Although these approaches have proven effective, they are limited to addressing only a single subtask within multimodal fake news detection, such as textual distortion, image distortion, or cross-modal distortion. Moreover, existing debate mechanisms typically operate at the object-level, where agents directly exchange evidence-based claims without explicit metacognitive reflection on their reasoning processes. A unified multi-agent debate framework capable of simultaneously handling multiple subtasks remains lacking.

3 Problem Formulation

Given a multimodal fake news dataset $\mathcal{D} = \{(m_i, y_i)\}_{i=1}^{N}$, each sample $m_i = (I_i, T_i)$ consists of an image I_i and its corresponding text T_i. While traditional methods treat fake news detection as a binary task, our MAR (Metacognitive Agentic Reasoning) framework addresses it as a fine-grained classification problem. The objective is to map the multimodal input to a label space $\mathcal{Y} = \{y_{\text{real}}, y_{\text{text}}, y_{\text{img}}, y_{\text{cross}}\}$, where y_i represents one of four categories: Real News, Textual Distortion, Visual Distortion, or Cross-modal Distortion. Instead of a simple binary mapping, our framework defines a unified multi-agent discriminative function F_{MAR} that maps the multimodal inputs directly to the four-class label space. For a given sample m_i, the final predicted label $\hat{y}_i$ is determined as:

$$\hat{y}_i, Exp = F_{\text{MAR}}(I_i, T_i; \Theta_{\text{MAR}}) \tag{1}$$

where Θ_{MAR} represents the parameters of the MAR framework. The Exp output is optional, providing a natural language explanation or a trace of the reasoning process. By shifting the focus to this four-class formulation, the MAR framework can explicitly reason over modality-specific inconsistencies and cross-modal semantic conflicts, providing a more interpretable diagnosis of why a news sample is deemed fraudulent.

4 Methodology

In this section, we propose a unified framework named MAR, which uses a multi-agent debate mechanism driven by metacognitive reasoning to detect multimodal fake news. We first present the overall architecture of the framework, followed by a detailed description of its four core modules: the domain-specific role generation module, the multi-agent metacognitive debate module, the multimodal external fact verification module, and the decision and explanation module. The coordinated integration of these modules enhances both the reliability and interpretability of fake news detection.

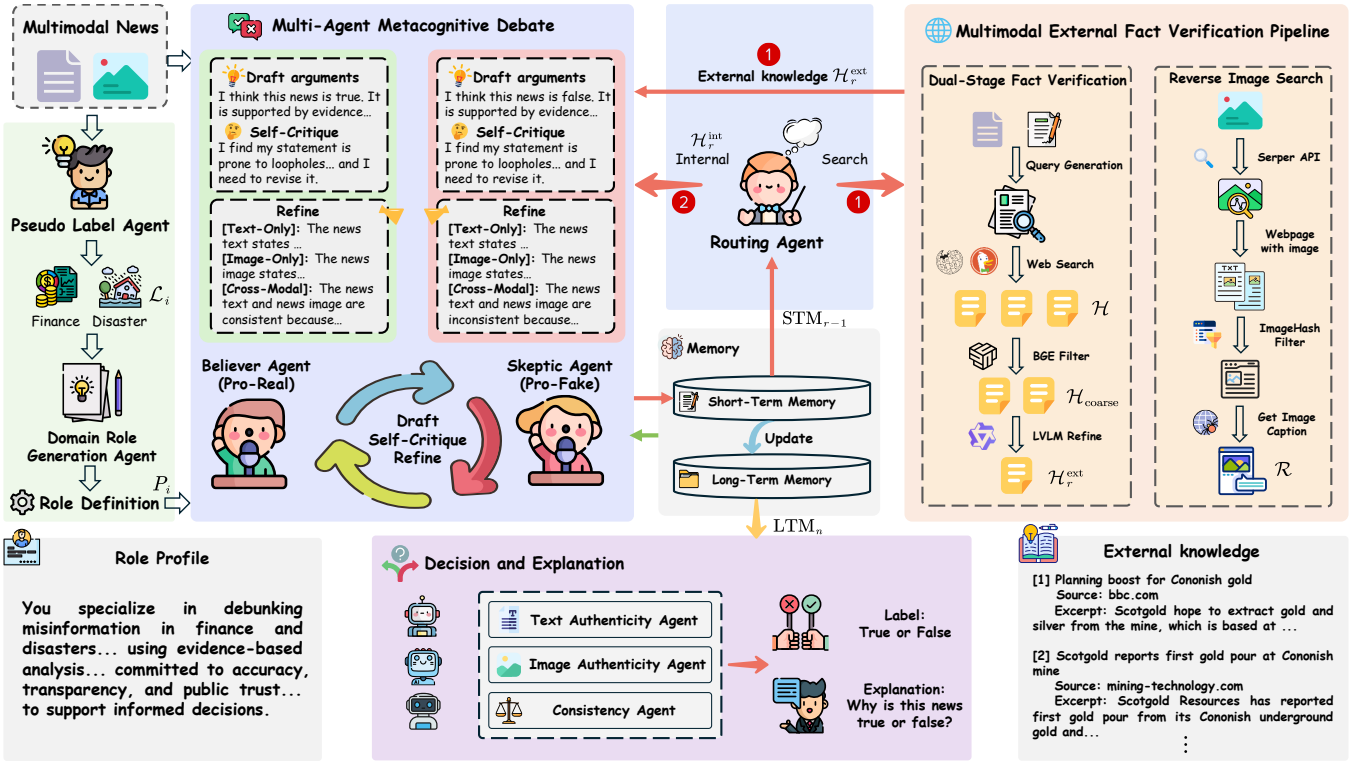


Figure 2: Illustration of the MAR framework. For multimodal news consisting of image–text pairs, MAR follows a multi-stage reasoning pipeline: (a) A Domain Role Generation Agent analyzes the news content and initializes domain roles. (b) Believer and Skeptic agents conduct a metacognitive debate, where a Routing Agent dynamically invokes external fact verification to retrieve cross-modal evidence and support the "Draft–Critique–Refine" cycle. (c) Finallythe Decision and Explanation module aggregates debate logs and verified evidence to produce the authenticity prediction and explanation.

4.1 The Overall Framework

The overall architecture of MAR is depicted in Figure 2. Given a multimodal news sample (I_i, T_i), MAR first analyzes the news content through a domain-specific role generation module to extract domain pseudo-labels and generate role definitions tailored to the domain, thereby incorporating domain knowledge into subsequent analysis. Next, the model introduces a metacognition-driven debate mechanism that constructs two opposing agents, Believer and Skeptic. To ensure a thorough and rigorous evaluation, the agents engage in three structured rounds of debate, simulating a deep cognitive cycle of "Draft, Self-Critique, and Refine" for in-depth assessment of information credibility. Before each round, a routing agent dynamically evaluates the debate state. If the agent detects a need for factual grounding such as specific entities, statistics, or unresolved contradictions, it triggers the Dual-Stage Fact Verification Engine for external web search to provide empirical evidence and suppress model hallucinations. In cases where such external evidence is not required, the system defaults to internal parametric reasoning to facilitate contextual understanding or general logical analysis. At the end of each round, the memory module first consolidates the raw debate logs into short-term memory (STM). This STM is then systematically integrated into the long-term memory (LTM), which is dynamically updated across rounds to maintain a coherent and evolving evidence chain. Finally, the system performs collaborative decision-making based on the accumulated LTM to output an explainable final judgment.

4.2 Basic Components

4.2.1 Domain-Specific Role Generation. To enhance the agents' specialized comprehension, we introduce domain-specific guidance through a two-stage process. First, a Pseudo Label Agent extracts a set of pseudo-labels $(\mathcal{L}_i)$ from the news text (T_i). To ensure precision, we constrain the number of labels to a maximum of three, allowing the system to capture the interdisciplinary nature of complex news while remaining focused on the most relevant themes. If no specific domains are identified, the sample is assigned to a "General Domain" by default.

Subsequently, the set $\mathcal{L}_i$ is passed to a Domain Role Generation Agent. Instead of generating separate roles for each label, this agent synthesizes the semantic features of all identified domains into a singular, cohesive Agent Persona P_i. This integrated identity provides the professional foundation for subsequent analysis, enabling the system to evaluate news from a multi-dimensional yet unified perspective.

4.2.2 Routing Agent. Assuming the system is currently at the r-th debate round, we introduce a Routing Agent to dynamically refine

the knowledge acquisition strategy prior to this round. This agent jointly analyzes the input news text T_i and the short-term memory STM_{r-1} to compute a Routing Confidence Score $D_r \in [0, 1]$. The knowledge source $\mathcal{K}_r$ is determined by a threshold-gated function:

$$\mathcal{K}_r = \begin{cases} \mathcal{H}_r^{\text{ext}}, & \text{if } D_r > \tau_{\text{route}} \\ \mathcal{H}_r^{\text{int}}, & \text{if } D_r \leq \tau_{\text{route}} \end{cases} \tag{2}$$

where τ_{route} denotes a predefined activation threshold, $\mathcal{H}_r^{\text{ext}}$ represents external evidentiary retrieval via web search, and $\mathcal{H}_r^{\text{int}}$ denotes internal parametric reasoning. The agent computes a high D_r and selects $\mathcal{H}_r^{\text{ext}}$ primarily when T_i or STM_{r-1} contains specific, verifiable entities such as names, dates, statistics, or time-sensitive official events, or when STM_{r-1} signals a factual impasse indicated by unresolved contradictions or explicit disagreements among debaters. In these cases, external retrieval supplies empirical evidence to support verification and resolve conflicts. Conversely, when the content is primarily subjective, opinion-based, or relies on general background knowledge without concrete verifiable anchors, the agent computes a low D_r and the system defaults to $\mathcal{H}_r^{\text{int}}$, relying exclusively on the internal parametric knowledge of the large language model for reasoning. This mechanism ensures that $\mathcal{K}_r$—the knowledge source selected for round r—is contextually optimized, achieving a synergy between factual precision and computational economy throughout the multi-round debate.

4.3 Multimodal External Fact Verification

Fake news detection is fundamentally a fact-checking task that demands up-to-date and verifiable knowledge. However, Large Language Models (LLMs) inherently suffer from persistent knowledge lag and hallucinations, making them ill-suited for standalone verification. To bridge this gap, this paper proposes a multimodal external fact-verification pipeline comprising a dual-stage verification engine alongside a reverse image search module. This design enables the framework to retrieve real-time factual evidence from heterogeneous sources seamlessly and integrate it directly into the reasoning process.

4.3.1 Dual-Stage Fact Verification Engine. This module is engineered to retrieve external knowledge and execute multi-stage filtering to guarantee the precision and relevance of verified information. The core pipeline integrates query generation, external web search, coarse-grained filtering powered by the bge-base-en-v1.5 embedding[44] model[2], and fine-grained refinement leveraging the Qwen3-VL-32B-Instruct-FP8 LVLM[5]. Through this cascaded filtering process, raw web results are systematically distilled into high-quality external evidence with enhanced relevance and factual reliability.

The process begins by synthesizing the textual information of the current news T_i with the dialogue logs from the previous debate round STM_{r-1} to construct the anchor context A_r:

$$A_r = \mathrm{Concat}(T_i, \mathrm{STM}_{r-1}) \tag{3}$$

Building upon this context, the specialized Query Generation Agent intelligently analyzes the information to generate a set of correlated search queries, denoted as $Q_r = \{q_1, q_2, \ldots, q_m\}$.

For each query $q_i \in Q_r$, the system utilizes the DuckDuckGo API[3] to perform web searches. The results H_i from all queries are aggregated into a raw retrieval set $\mathcal{H}$. For each individual result h_j, a textual representation s_j is constructed by concatenating its title and excerpt:

$$\begin{aligned} \mathcal{H} &= \bigcup_{i=1}^{m} H_i = \{h_1, h_2, \ldots, h_n\} \\ s_j &= \mathrm{Concat}(\mathrm{title}_j, \mathrm{excerpt}_j) \end{aligned} \tag{4}$$

To mitigate noise inherent in direct web retrieval, this research employs a two-step refinement process. First, the BGE embedding model is used for coarse filtering. The system computes the cosine similarity between the embedding of the anchor context A_r and each search result s_j, selecting the top-k most relevant results to form the set $\mathcal{H}_{\text{coarse}}$:

$$\begin{aligned} \mathbf{a} = \mathrm{BGE}(A_r), \quad \mathbf{s}_j &= \mathrm{BGE}(s_j) \\ \mathrm{sim}(\mathbf{a}, \mathbf{s}_j) &= \frac{\mathbf{a}^\top \mathbf{s}_j}{\|\mathbf{a}\| \|\mathbf{s}_j\|} \\ \mathcal{H}_{\text{coarse}} &= \arg\max_{\substack{S \subseteq \mathcal{H} \\ |S|=k}} \sum_{h_j \in S} \mathrm{sim}(\mathbf{a}, \mathbf{s}_j) \end{aligned} \tag{5}$$

For each candidate $h_j \in \mathcal{H}_{\text{coarse}}$, the Qwen3-VL-32B-FP8 model is employed to perform a joint assessment of relevance and credibility. By scoring the results against the anchor context A_r and applying a confidence threshold τ, the system filters out unreliable information to complete the dual-stage verification, where $\mathcal{H}_r^{\text{ext}}$ denotes the final output of this module:

$$\begin{aligned} \mathrm{score}_j &= \mathrm{LVLMScore}(A_r, h_j) \\ \mathcal{H}_r^{\text{ext}} &= \left\{ h_j \in \mathcal{H}_{\text{coarse}} \mid \mathrm{score}_j \geq \tau \right\} \end{aligned} \tag{6}$$

4.3.2 Reverse Image Search. The module utilizes the Serper API[4] to perform reverse image searches on a given news image I_i in order to retrieve web-based information related to the image. This process gathers a collection of initial matching results $\mathcal{M}^{\text{raw}} = \{m_j\}_{j=1}^{M}$ ($M \leq 20$), where each result m_j consists of the source domain, the news page URL (u_j^{page}), and the corresponding image URL (u_j^{img}).

To filter out search results that deviate significantly from the original image, the module applies Perceptual Hashing (pHash) to compare the retrieved image I_j^{web} with I_i by calculating the Hamming distance:

$$d_j = \mathrm{HammingDist}(\mathrm{pHash}(I_i), \mathrm{pHash}(I_j^{\text{web}})) \tag{7}$$

To ensure visual consistency between the retrieved images and the original image, a similarity threshold δ_{re} is applied to filter the candidates. For images meeting this threshold, the module extracts the corresponding descriptive text S_j (caption) by deploying a crawler $\mathcal{F}(\cdot)$ to parse the HTML source of u_j^{page}. The final result set $\mathcal{R}$ is then constructed by aggregating the top N validated captions:

$$\mathcal{R} = \{(m_j, S_j) \mid d_j \leq \delta_{\text{re}}, S_j = \mathcal{F}(u_j^{\text{page}})\}_{j=1}^{N} \tag{8}$$

where $N \leq M$ denotes the number of results passing the visual similarity verification.

[2] https://huggingface.co/BAAI/bge-base-en-v1.5

[3] https://github.com/deedy5/ddgs

[4] https://google.serper.dev/lens

4.4 Multi-Agent Metacognitive Debate

The Metacognitive Stage serves as the core of the MAR framework, aiming to evaluate news authenticity and perform attribution diagnosis to identify specific forgery patterns through multimodal debates. This stage establishes two opposing roles, namely the Believer and the Skeptic. Since multimodal fact-checking is highly susceptible to unproductive circular arguments, we introduce a metacognitive constraint mechanism.

Inspired by the proverb "rumors stop at the wise," we simulate real-world debating logic by designing a three-stage rational reasoning model that consists of Draft, Self-Critique, and Refine. Crucially, we integrate external knowledge into this process to facilitate fact-based verification and mitigate potential hallucinations. Through this closed-loop process of proposal, evidence-based reflection, and distillation, the framework constructs a "wise" entity endowed with rational analytical capabilities, thereby ensuring that debates converge toward rigorous logical depth and factual accuracy. The systematic workflow of this three-stage reasoning model is formalized through the following stages:

4.4.1 Draft Stage. In the initial phase, the Believer and Skeptic agents analyze the multimodal sample $m_i = (T_i, I_i)$. Guided by their persona definitions P_i and explicit stance orientations $S \in \{S_{true}, S_{fake}\}$, they leverage the knowledge source($\mathcal{K}_r$), reverse image search results ($\mathcal{R}$), and historical memory (consisting of short-term controversies STM_{r-1} and long-term context LTM_{r-1}) to formulate a preliminary stance D_r. Specifically, $\mathcal{K}_r$ is dynamically determined by a Routing Agent, which selects either the LLM's internal parametric knowledge or real-time information retrieved via Dual-Stage Fact Verification. S_{true} directs the Believer to construct arguments supporting the news's authenticity, while S_{fake} prompts the Skeptic to identify potential deceptive patterns. The process is formalized separately for each agent:

$$\begin{aligned} D_r^B &= \text{Draft}(P_i, m_i, \text{LTM}_{r-1}, \text{STM}_{r-1}, \mathcal{K}_r, \mathcal{R}, S_{true}) \\ D_r^S &= \text{Draft}(P_i, m_i, \text{LTM}_{r-1}, \text{STM}_{r-1}, \mathcal{K}_r, \mathcal{R}, S_{fake}) \end{aligned} \tag{9}$$

where $\mathcal{R}$ specifically denotes the evidence retrieved from reverse image searches to verify the original context and source of the visual content.

4.4.2 Self-Critique Stage. To ensure logical rigor and avoid leaving vulnerabilities for the opponent to exploit, agents perform an internal audit of D_r. By re-examining the evidence, agents identify and eliminate subjective conjectures or claims lacking evidentiary support. This introspective process yields self-corrected viewpoints:

$$\begin{aligned} C_r^B &= \text{SelfCritique}(m_i, D_r^B, \mathcal{K}_r, \mathcal{R}) \\ C_r^S &= \text{SelfCritique}(m_i, D_r^S, \mathcal{K}_r, \mathcal{R}) \end{aligned} \tag{10}$$

4.4.3 Refine Stage. In the final phase, agents integrate the cognitive outputs of the previous stages (D_r, C_r) while adhering to their role definitions P_i to produce a polished rebuttal or reinforcement R_r. To mitigate hallucinations and ensure robust reasoning, this stage incorporates grounded references: $\mathcal{K}_r$, a hybrid knowledge source dynamically routed by the Routing Agent to select between the LLM's internal parametric knowledge and real-time evidence retrieved via Dual-Stage Fact Verification, together with $\mathcal{R}$ obtained through reverse image search. These complementary knowledge streams ensure that the final rebuttal or reinforcement is both precise and evidence-backed:

$$\begin{aligned} R_r^B &= \text{Refine}(P_i, m_i, D_r^B, C_r^B, \mathcal{K}_r, \mathcal{R}, S_{true}) \\ R_r^S &= \text{Refine}(P_i, m_i, D_r^S, C_r^S, \mathcal{K}_r, \mathcal{R}, S_{fake}) \end{aligned} \tag{11}$$

By adopting this structured pipeline, the agents effectively evaluate multimodal fake news across three critical dimensions: text-specific veracity, image-specific veracity, and cross-modal consistency. This comprehensive approach enables the "wise" agents to better utilize multi-source evidence to substantiate their final viewpoints.

4.4.4 Contextual Memory. To mitigate the adverse effects of escalating debate logs on the inference efficiency and contextual coherence of Large Language Models (LLMs), this paper employs a Dual-Memory System framework. This mechanism first distills the raw arguments into a Short-Term Memory (STM_r) for the current round by extracting core controversies and consensus between the debating parties. Subsequently, the system adopts a recursive update strategy to fuse the current short-term memory with the preceding Long-Term Memory (LTM_{r-1}). This process constructs a dynamically evolving long-term memory (LTM_r) structure, ensuring the persistence and consistency of key information throughout long-range dialogues.

$$\begin{aligned} \text{STM}_r &= \text{Summarize}(R_r^B, R_r^S) \\ \text{LTM}_r &= \text{Update}(\text{LTM}_{r-1}, \text{STM}_r) \end{aligned} \tag{12}$$

Here, R_r^B and R_r^S represent the raw responses of the two debating parties in round r. The Summarize function is implemented by a summarization agent to distill core information, while the Update function is performed by a memory management agent to integrate the short-term memory into the long-term repository.

4.5 Decision and Explanation

Upon completion of all debate rounds, the system retrieves information from the final long-term memory LTM_n, specifically targeting the debate logs and consensus outcomes for the set of key dimensions $\mathcal{D} = \{\text{text}, \text{visual}, \text{consistency}\}$. Three specialized agents, each aligned with a specific dimension $d \in \mathcal{D}$, then perform in-depth analyses by synthesizing both the deliberation process captured in the logs and the final consensus. These agents translate their analyses into structured rationales E_d and category-level judgments L_d. Finally, the system integrates these agent outputs to generate the definitive assessment of the multimodal news item, consisting of a final classification label L_{final} and an integrated explanatory rationale E_{final}:

$$\begin{aligned} \mathcal{D} &= \{\text{text}, \text{visual}, \text{consistency}\} \\ (L_{final}, E_{final}) &= \Phi\Big(\{(L_d, E_d)\}_{d \in \mathcal{D}}\Big) \end{aligned} \tag{13}$$

5 Experiments

5.1 Experimental Setup

5.1.1 Dataset. Our experiments utilize two multimodal datasets: MMFakeBench[24], which contains 11,000 samples across four categories (Real News, Textual Veracity Distortion, Visual Veracity Distortion, and Image-Text Consistency Distortion), and DGM^4[32],

Table 1: The performance of MAR(Ours), SVLM-only(G1), LVLM-only(G2), and the Multi-Agent System(G3) approach was evaluated on the binary classification task.

Group	Model	MMFakeBench				DGM^4			
		F1	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall
G1	MCAN[43]	67.26	71.60	66.85	67.90	52.12	56.60	52.47	52.81
	EARAM[48]	75.08	79.50	75.68	74.60	61.07	68.50	61.63	60.74
G2	Qwen2.5-VL-32B	66.39	68.20	66.99	70.14	66.36	68.90	66.27	68.93
	Qwen3-VL-32B	66.09	76.20	73.72	64.71	58.44	73.40	70.56	58.81
	GPT-4o	75.67	77.30	75.29	**79.69**	66.56	67.60	68.60	72.10
	GPT-4o-mini	70.79	78.30	75.71	69.07	57.42	72.90	69.35	58.07
	GLM-4.6V	72.27	73.70	72.69	76.93	68.70	69.70	70.58	74.45
G3	LRQ-FACT(w/o RAG)[8]	67.34	72.70	67.43	67.26	61.89	67.90	61.86	61.93
	LRQ-FACT(w/ RAG)[8]	70.22	74.50	69.89	70.64	62.71	66.70	62.42	63.83
	MMD-Agent[24]	74.38	79.50	75.95	73.36	69.66	73.70	69.19	70.36
Ours	MAR	**78.05**	**81.30**	**77.69**	78.45	**72.75**	**75.70**	**72.00**	**74.45**

which comprises 230,000 news samples. For the latter, we remapped the original DGM^4 labels to align with our task as follows[8]: Real (from original), Textual Veracity Distortion (from Text Attribute), Visual Veracity Distortion (from Face Swap and Face Attribute), and Image-Text Consistency Distortion (from Text Swap).

5.1.2 Baselines. The current multimodal fake news detection methods mainly include those based on small language models, those based on large vision language models, and those based on multi-agent systems. We compare three groups of methods.

G1 (SVLM-Only): We compared two small-scale vision-language models as our baselines. **1) MCAN**[43]: This model jointly models the spatial and frequency domain features of an image. It utilizes a multi-layer co-attention mechanism to achieve a deep fusion of textual and visual information. **2) EARAM**[48]: This model leverages a Large Vision-Language Model (LVLM) to generate commonsense reasoning and logical analysis of the image-text pair. Subsequently, a Multimodal Small Language Model (MSLM) adaptively extracts useful rationales to make a final judgment and guides the LVLM in generating an explanation.

G2 (LVLM-Only): We selected five advanced Large Vision Language Models (LVLMs) as baselines: Qwen2.5-VL-32B[6], Qwen3-VL-32B [5], GPT-4o [19], GPT-4o-mini (a lightweight variant of GPT-4o), and GLM-4.6V [16]. To ensure a fair and rigorous comparison, all experiments were conducted under a zero-shot setting using consistent standard prompts across all models. This setup aims to evaluate the intrinsic multimodal reasoning and comprehension capabilities of each model without the influence of external tools or complex prompt engineering.

G3 (Multi-Agent System): We selected two multi-agent systems focused on fine-grained misinformation identification as our baselines: **1) MMD-Agent[24]**: This system detects multimodal fake news from mixed sources via staged reasoning and external Wikipedia knowledge retrieval. **2) FRQ-FACT[8]**: This system leverages large vision-language models (LVLMs) to automatically generate fact-checking questions (FCQs) for image-text content. To comprehensively evaluate its performance, we compare two versions: LRQ-FACT (w/o RAG), which answers FCQs based solely on the model's internal knowledge, and LRQ-FACT (w/ RAG), which integrates textual Q&A with retrieved external evidence to verify claims.

5.1.3 Implementation Details. Following the original 3:3:1:3 category distribution of MMFakeBench, we constructed a stratified subset of 6,000 samples, partitioned into 5,000 training and 1,000 test instances (a 5:1 split). Specifically, the training set comprises 1,500 Real News, 1,500 Textual Veracity Distortion, 500 Visual Veracity Distortion, and 1,500 Image-Text Consistency Distortion samples; the test set contains 300, 300, 100, and 300 samples from each category, respectively. For DGM^4, we remapped its original labels to this four-category framework using the aforementioned strategy and then applied identical stratified sampling to obtain 5,000 training and 1,000 test samples while preserving the 3:3:1:3 ratio.

For SLM-based baselines, we trained models on the 5,000-sample training set using their default best-practice hyperparameters and evaluated them on the test set. For LLM-based methods, we evaluated models directly on the test set without training. Across all experiments, the temperature parameter was uniformly set to 0 for all large language models to ensure deterministic outputs and reproducible results. In our proposed MAR, the multi-agent system undergoes three rounds of debate governed by a metacognitive reasoning framework; this structure allows the agents to not only exchange information but also to monitor their own reasoning gaps and iteratively refine their logic to balance cognitive depth and computational efficiency. Unless otherwise specified, Qwen3-VL-32B-Instruct-FP8[5] served as the backbone model.

5.2 Performance of Our MAR Model

We evaluate the performance of MAR on both binary and multiclass classification tasks, using Macro-F1, Accuracy, Precision, and Recall as evaluation metrics.

[5]https://huggingface.co/Qwen/Qwen3-VL-32B-Instruct-FP8

Table 2: The performance of MAR(Ours), SVLM-only(G1), LVLM-only(G2), and the Multi-Agent System(G3) approach was evaluated on the multi-class classification task.

Group	Model	MMFakeBench				DGM^4			
		F1	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall
G1	MCAN[43]	40.93	49.60	44.87	43.17	27.53	34.90	28.75	29.08
	EARAM[48]	67.26	67.20	69.29	66.83	47.07	52.00	47.33	47.67
G2	Qwen2.5-VL-32B	43.48	49.70	54.80	44.58	31.88	46.90	34.46	39.08
	Qwen3-VL-32B	40.07	49.40	56.19	41.67	28.81	41.10	41.97	34.25
	GPT-4o	45.28	58.20	49.82	48.50	37.28	53.00	41.95	44.17
	GPT-4o-mini	33.33	45.30	50.72	37.75	22.04	37.10	47.32	30.92
	GLM-4.6V	45.36	52.20	46.85	47.33	42.61	52.40	47.19	44.33
G3	LRQ-FACT(w/o RAG)[8]	57.50	58.10	62.74	59.42	34.19	44.40	42.86	37.33
	LRQ-FACT(w/ RAG)[8]	60.80	61.50	64.76	62.42	38.16	49.00	47.05	41.17
	MMD-Agent[24]	52.54	55.10	53.42	56.08	43.39	50.60	43.45	43.50
Ours	MAR	**68.02**	**68.20**	**69.62**	**69.33**	**47.74**	**60.80**	**48.11**	**50.67**

Table 3: Performance Comparison of Ablation Variants on Binary Classification

Variants	MMFakeBench		DGM^4	
	F1	Acc.	F1	Acc.
MAR	**78.05**	**81.30**	**72.75**	**75.70**
w/o Domain	77.48	81.10	71.87	75.60
w/o Metacognitive	77.63	81.00	71.86	74.60
w/o MAMD	76.09	79.10	71.44	73.30
w/o Reverse	75.80	79.50	71.24	74.00
w/o MEFV	71.61	77.98	62.29	72.80

5.2.1 Binary Classification Tasks. Table 1 compares the proposed MAR framework with representative models on multimodal fake news detection (binary classification). Key findings: **1)** MAR achieves state-of-the-art (SOTA) performance across most metrics on both MMFakeBench and DGM^4. It reaches the highest F1 and Accuracy, demonstrating its superior capability in identifying complex cross-modal conflicts. **2)** MAR substantially outperforms SVLM-only methods (G1), improving F1 scores by 2.97% and 11.68% over the strongest baseline (EARAM), highlighting its effectiveness in capturing subtle image-text contradictions that traditional models miss. **3)** MAR outperforms all LVLM-only baselines (G2) in overall metrics (F1 and Accuracy), including state-of-the-art models such as GLM-4.6V and GPT-4o. This demonstrates that MAR's multi-agent collaborative reasoning overcomes the limitations of individual models, enabling more effective multimodal evidence synthesis than single models using standard prompting. **4)** Among multi-agent architectures (G3), MAR outperforms MMD-Agent by 3.67% and 3.09% in F1 score across the two datasets, demonstrating the effectiveness of its metacognitive reasoning in agent collaboration and decision fusion for detecting highly concealed fake news.

5.2.2 Multi-Class Classification Tasks. Table 2 evaluates the proposed MAR framework against representative models on multi-class fake news detection (distinguishing text veracity distortion, image veracity distortion, and text-image consistency distortion) using MMFakeBench and DGM^4. Key results: **1)** MAR achieves SOTA performance across both datasets, with F1 scores of 68.02% and 47.74%. By leveraging a metacognitive debate mechanism, it effectively identifies fine-grained forgery types—including text, image, and cross-modal manipulations—outperforming all baselines in both F1 and precision. **2)** MAR improves over SVLM-only methods (G1) by 0.76% and 0.67% in F1 by introducing explicit cross-modal agent interaction, which overcomes G1's limitation of isolated modality processing and captures subtle cross-modal contradictions missed by single-branch architectures. **3)** MAR surpasses all LVLM-only baselines (G2) in F1 score, suggesting that the fixed reasoning pathways of individual models often struggle with the complexity of multi-class forgery attribution. In contrast, MAR's multi-agent debate enables iterative hypothesis refinement grounded in evidence, leading to more robust cross-modal alignment and superior categorization compared to standard zero-shot inference. **4)** Compared to the multi-agent baseline LRQ-FACT(w/ RAG) (G3), MAR gains 7.22% and 9.58% in F1, demonstrating that meta-cognitive reasoning integrated with multi-agent debate facilitates more effective identification of fine-grained multimodal fake news categories.

5.3 Ablation Study

In the ablation study, we conduct experiments on the binary classification task to investigate the contribution of each component within the MAR model. By systematically removing specific modules, we evaluate the following five variants: **1) w/o Domain:** Removes the persona profiles generated via news domain pseudo-labels. **2) w/o Metacognitive:** Eliminates the metacognitive reasoning process within the multi-agent debate by removing the Self-Critique and Refine stages, retaining only the Draft stage. **3) w/o MAMD:** Removes the entire Multi-Agent Metacognitive Debate (MAMD) component, instead directly utilizing the multimodal news combined with external knowledge for final decision-making. **4) w/o Reverse:** Excludes the caption information obtained through

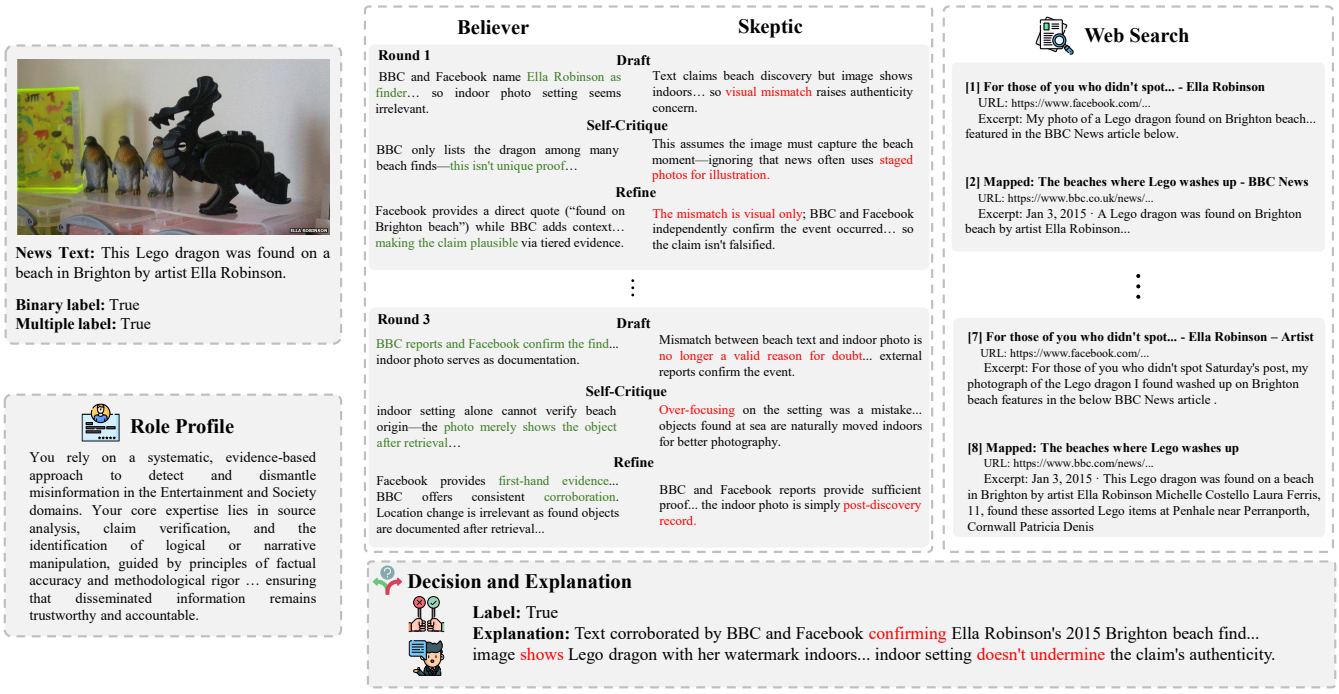


Figure 3: Multimodal News Detection and Explanation for True News Using the MAR Framework

reverse image searching of news images. **5) w/o MEFV:** Removes the entire Multimodal External Fact Verification (MEFV) component; in this configuration, the model does not incorporate external knowledge and relies solely on the LLM's internal knowledge for the multi-agent metacognitive debate.

The experimental results appear in Table 3. The full MAR model consistently outperforms all ablation variants across both datasets, confirming the synergy among its components. Removal of the MEFV module causes the largest performance decline, with over 10% reduction in F1 score on DGM^4, underscoring external factual evidence as a cornerstone for detecting sophisticated misinformation. Disabling metacognitive reasoning or the MAMD mechanism also yields substantial accuracy losses. These gaps indicate that the multi-agent debate process, particularly the "self-critique and refine" loop, effectively emulates human-like higher-order reasoning to enhance detection performance.

5.4 Case Study

To clearly demonstrate the reasoning process of the MAR model, we selected a representative case from the MMFakeBench dataset (see Figure 3). This case centers on a news report about "Robinson finding a Lego dragon on Brighton beach." Initially, Believer and Skeptic held opposing views: Believer, drawing on external retrieval information, argued that the event was authentic and that the indoor photography setting did not affect factual determination; Skeptic, however, questioned the visual mismatch between the beach discovery described in the text and the indoor environment shown in the image, raising concerns about authenticity. Through multi-round debate and self-reflection, both parties gradually revised their initial judgments and ultimately reached a consensus: the indoor photograph merely documented the object after retrieval, and the location change was a reasonable practice that did not constitute a factual contradiction. Thus, the news report was confirmed as authentic.

6 Conclusion

In this paper, we propose MAR, a novel framework that integrates metacognitive multi-agent debate with dynamic external knowledge retrieval for multimodal fake news detection. MAR introduces a domain-informed reasoning mechanism that replicates human-like metacognitive patterns through a structured "Draft–Self-Critique –Refine" workflow. By using pseudo-labels to synthesize specialized role profiles and incorporating real-time evidence from the web and reverse image searches, MAR effectively mitigates model hallucinations while ensuring high factual accuracy. Extensive experiments on two benchmark datasets demonstrate that MAR achieves state-of-the-art performance, showcasing exceptional generalization and inferential rigor in complex cross-modal scenarios.

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A Evaluations on Explanation

Existing automatic evaluation mechanisms heavily rely on large language models (LLMs) as arbiters, yet their reliability is often compromised by two critical biases: positional bias[39], where the ordering of candidate responses unduly influences evaluation outcomes; and self-preference bias[35], where models tend to favor responses generated by architectures from the same family. These biases may undermine the objectivity of automated assessments. To mitigate these limitations, this paper adopts a rigorous human evaluation protocol, employing a 5-point Likert scale (1 = Very Poor, 5 = Excellent) to quantitatively assess the quality of model-generated explanations across the following five dimensions:

- **Precision**: Evaluates whether the explanation accurately identifies and articulates key evidence from multimodal inputs. High-scoring explanations pinpoint concrete contradictions—such as specific entities, geographic inconsistencies, temporal discrepancies, or visual artifacts indicative of manipulation—rather than resorting to vague assertions like "the information is unreliable."
- **Neutrality**: Assesses the neutrality and empiricism of the explanation. High-scoring explanations rely solely on logical reasoning and verifiable factual evidence, strictly avoiding emotive language, subjective conjectures, or unsupported speculations.
- **Consistency**: Measures the logical coherence between the explanation and the predicted label. The explanation must provide a clear rationale that directly justifies the assigned category—true news, text veracity distortion, image veracity distortion, or text-image consistency distortion—without contradictions between the reasoning and the final judgment.
- **Readability** : Evaluates the textual quality and comprehensibility of the explanation. High-scoring explanations exhibit clear structure, grammatical correctness, and conciseness, enabling non-expert readers to rapidly grasp the core reasoning behind the classification decision.
- **Synergy**: Assesses the model's capacity to synthesize information across modalities. High-scoring explanations demonstrate joint analysis of both textual details and visual features, explicitly articulating their complementary or conflicting relationships to support the final judgment, rather than relying predominantly on a single modality.

To implement this protocol, we randomly selected 50 samples from the MMFakebench dataset for a detailed evaluation of their explanatory rationales. We recruited 10 volunteers to perform the assessment. For a comparative analysis, we also evaluated the explanations generated by Qwen3-VL-32B and GPT-4o-mini under standard prompting conditions. The comprehensive results of this human evaluation are illustrated in Figure 4. Overall, our proposed MAR model outperforms both Qwen3-VL-32B and GPT-4o-mini in four out of the five evaluated dimensions. As shown in the radar chart, MAR exhibits significant advantages in Precision and Neutrality, which demonstrates a superior ability to pinpoint factual contradictions while maintaining an objective stance. Furthermore, the model shows stronger Synergy and Consistency, reflecting its enhanced capacity to integrate textual and visual evidence into a coherent rationale. While the baselines maintain a slight lead in Readability, the overall results validate that MAR provides more reliable and logically rigorous explanations than the state-of-the-art models.

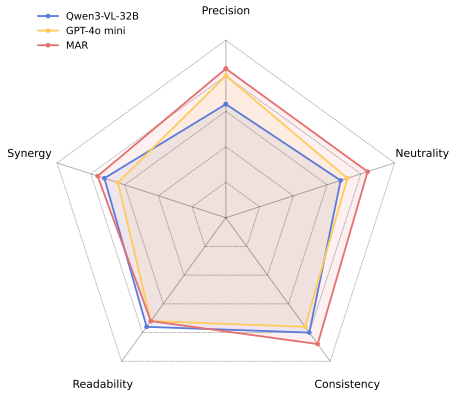


Figure 4: Interpretability evaluation of our model on the MMFakebench dataset.

B Retrieve Failed Case Study

We further evaluated the model's performance under retrieval failure scenarios, where the Routing Agent determined that the LLM's internal knowledge alone suffices for veracity assessment, as illustrated in the case study of Figure 5. We found that even without external knowledge, after deliberation through our MAR framework, both Believer and Skeptic agents reached a consensus: while the news image depicts adaptive optics solely in its astronomical context, the text accurately extends to broader cross-disciplinary applications. The MAR judgment concluded that the image serves as a contextual illustration of the technology's origin, while the text expands on its wider scientific utility—these modalities are complementary rather than contradictory, supporting the overall truthfulness of the news claim.

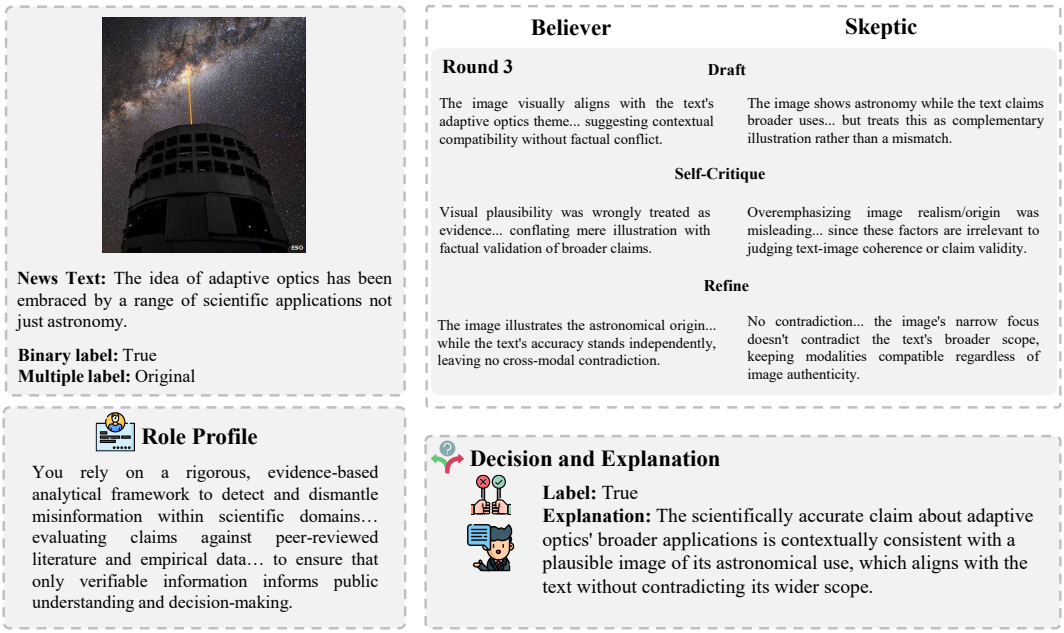


Figure 5: Retrieve Failed Case Study

C Cost and Runtime Analysis

We further evaluated the resource consumption of our proposed MAR model under a multi-threaded inference setting on the MMFakeBench dataset. Since MAR leverages vLLM for local LVLM deployment, no commercial token cost is incurred during actual inference, rendering the direct monetary cost negligible. To estimate the potential expense under a commercial API scenario, we referenced the official pricing of Qwen3-VL-32B-Instruct ($0.16 per million input tokens and $0.64 per million output tokens). Under this setup, the model consumes an average of 60,591.13 tokens per sample. Based on the referenced pricing, this translates to an estimated commercial cost of $0.01515 per sample, with an average inference latency of 21.79 seconds.

D Ethical Considerations

The MAR framework aims to limit the spread of multimodal misinformation and reduce its negative societal impact, especially in sensitive areas like public opinion and social stability. While the framework seeks to strengthen public resilience against misinformation, we acknowledge that malicious actors might attempt to exploit or bypass such systems. We oppose any misuse of this technology and support its responsible deployment in human-centered systems, with human oversight to ensure robustness, accountability, and transparency.

We also prioritize the well-being and dignity of research participants. Ten volunteers took part in our human evaluation after receiving a clear explanation of the study's purpose and potential exposure to misleading content. They were informed they could pause or withdraw at any time without penalty. To protect psychological well-being, sessions were kept under 30 minutes, participants were encouraged to take breaks as needed, and content depicting graphic violence or targeted harassment was excluded. Mental health resources were provided, and all participants received fair compensation.

On data ethics, our data use was limited to publicly available news content and conducted strictly for academic research purposes. Any personally identifiable information—such as names of ordinary individuals—was removed during preprocessing, ensuring no private user data is retained in the MMFakeBench or DGM[4] datasets. References to public figures in news reports were preserved as part of the public discourse context. The datasets and associated modules are intended solely for research on multimodal misinformation detection; any use for surveillance, targeting individuals, or other non-research purposes is strictly prohibited. All data handling complies with applicable laws, platform terms, and institutional ethical guidelines.